

Practical 1

This practical will enable you to think about key considerations relating to dataset composition that are important when we think about questions of fairness.

COMPAS Case study data exploration

In this practical you will get to know more about the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) and the ProPublica analysis discussed in the lectures. This data was used to make predictions about the risk that a criminal defendant would re-offend.

The goal of this case study is to explain the statistical arguments presented in two articles from 2016:

- "[Machine Bias](#)", by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, and published by [ProPublica](#). Also see: [How We Analyzed the COMPAS Recidivism Algorithm](#).
- A response by Sam Corbett-Davies, Emma Pierson, Avi Feller and Sharad Goel: "[A computer program used for bail and sentencing decisions was labelled biased against blacks. It's actually not that clear.](#)", published in the Washington Post.

I recommend reading these articles first which provide context and information about the analysis.

1. ProPublica claims that COMPAS is unfair because "among defendants who ultimately did not reoffend, Blacks were more than twice as likely as Whites to be classified as medium or high risk."
2. Northpointe claims that COMPAS is fair because "scores mean essentially the same thing regardless of the defendant's race. For example, among defendants who scored a seven on the COMPAS scale, 60 percent of White defendants reoffended, which is nearly identical to the 61 percent of Black defendants who reoffended."

Steps

1. Download the COMPAS (Propublica recidivism) dataset, <https://github.com/propublica/compas-analysis>
2. Have a look at the dataset files.
 - a. Does it need any cleaning? Do you find any null, unrelated or missed data?

- b. What categorical data is included, do the demographic categories vary across datasets (e.g. compass-scores-raw vs two-year etc)?
- c. Can you think of data that might be useful but that is not included?

3. Split the dataset into different demographic groups (i.e. gender, ethnicity, etc.).

For each demographic subgroup, explore the following:

- compute the size of the groups,
- the average value for each (numeric) feature,
- the variance of each (numeric) feature,
- Do you observe any interesting differences between the different subgroups' statistics?

4. You can implement the following code and explore a little further. Follow the steps described in [this blog post](#).

There are additional suggested steps provided in the starter notebook on blackboard.

Go further

- a. See the ProPublica analysis of the data at <https://github.com/propublica/compass-analysis/blob/master/Compass%20Analysis.ipynb>
- b. Northpoint's defense was that the predicted scores were the same thing regardless of group – this is Calibration. This is demonstrated nicely in this [Colab notebook](#)

5. Discussion

in groups with those sitting near you, discuss any observations you have made.

- if you have observed any sort of bias, please describe it and explain the reasons why this bias has happened, from your point of view.

Things to consider:

- Who are the affected people?
- How are their views represented?
- What is fair, and according to who?
- Is it possible to reconcile different stakeholder perspectives (i.e. citizens who want safety, people who don't want to be falsely imprisoned, police officers who good arrest rates, etc.)?
- What goes into the risk score?
- How is it calculated?
- What is the accuracy?
- How is this accuracy measured?
- What definition of fairness was used to develop the scores?
- Why this definition instead of other definitions?
- What are potential fairness violations?

